

# MINING QUALITY CONTROL & PREDICTION SYSTEM

## Professional Project Documentation

CSV-Free Streamlit Prototype  
Independent Engineering / Software Project

| Item         | Description                                                       |
|--------------|-------------------------------------------------------------------|
| Project Type | Process-quality estimation and diagnostic dashboard               |
| Application  | Mining / mineral-processing process monitoring prototype          |
| Interface    | Streamlit web application                                         |
| Inputs       | pH, slurry intensity, reagent dose, feed rate                     |
| Output       | Estimated mineral-quality score, grade status and recommendations |

**Important scope statement:** This is a prototype estimator, not a laboratory instrument. The current targets, tolerances and weights are configurable assumptions and must be calibrated and validated against real plant/laboratory measurements before operational use.

# 1. Executive Summary

The Mining Quality Control & Prediction System is a CSV-free prototype designed to demonstrate how four process parameters can be converted into an easy-to-understand quality indicator. The application provides an interactive Streamlit dashboard in which a user can adjust pH, slurry intensity, reagent dose and feed rate, run an analysis, and receive an estimated mineral-quality percentage together with parameter-level scores and diagnostic recommendations.

The project translates process-engineering concepts into a usable software tool. Instead of requiring historical data or a machine-learning model, the current prototype uses a transparent weighted scoring method. This makes the calculation easy to explain during an interview and easy to calibrate when validated engineering data become available.

# 2. Problem Statement

Mining and mineral-processing operations depend on multiple process variables that influence product quality. Engineers may need to understand how changes in operating conditions affect a quality indicator. This prototype provides an interactive foundation for that type of process monitoring.

- Create a user-friendly interface for changing key process parameters.
- Convert deviations from desired operating regions into interpretable scores.
- Combine individual scores into one overall quality indicator.
- Provide diagnostic feedback rather than only displaying a single number.
- Create a foundation that can later be calibrated using laboratory or plant data.

# 3. Project Objectives

- Develop an interactive mining-quality dashboard using Streamlit.
- Allow users to adjust four process parameters without editing code.
- Calculate an estimated quality score from 0–100%.
- Show individual parameter contributions and target values.
- Generate basic diagnostic recommendations.
- Keep the first prototype CSV-free and easy to demonstrate.
- Provide an architecture that can later be calibrated with validated data.

# 4. Solution Workflow

1. User adjusts pH, slurry intensity, reagent dose and feed rate.
2. Each value is compared with a prototype optimum and tolerance.
3. A closeness function converts each parameter into a 0–100% score.
4. The four scores are combined using predefined weights.
5. The result is classified into a grade/status category.
6. The dashboard displays the result, parameter scores, targets and recommendations.

## 5. System Architecture

| Layer             | Function                                                              |
|-------------------|-----------------------------------------------------------------------|
| User Interface    | Streamlit sliders, metrics, tables, progress bars and analysis button |
| Calculation Layer | Closeness-score function and weighted quality calculation             |
| Diagnostic Layer  | Grade classification and parameter recommendations                    |
| Output Layer      | Estimated quality %, parameter scores, targets and status             |

## 6. Input Parameters & Prototype Configuration

| Parameter        | Optimum     | Tolerance | Weight | UI Range        |
|------------------|-------------|-----------|--------|-----------------|
| pH               | 8.0         | 0.9       | 30%    | 5.0–11.0        |
| Slurry intensity | 50.0        | 12.0      | 20%    | 30.0–70.0       |
| Reagent dose     | 18.0 mL     | 6.0       | 25%    | 5.0–30.0 mL     |
| Feed rate        | 220 tons/hr | 25.0      | 25%    | 170–270 tons/hr |

These are prototype configuration values. They should not be presented as validated operating limits for a specific mineral-processing plant.

## 7. Calculation Method

### 7.1 Individual Parameter Score

The application measures how far the current value is from the selected optimum. The deviation is normalized by the tolerance and passed through a Gaussian-like closeness function:

$$\text{Score} = 100 \times \exp(-(d^2 / 2)), \text{ where } d = (\text{current} - \text{optimum}) / \text{tolerance}$$

A value exactly at the optimum produces 100%. The score decreases smoothly as the value moves farther away.

### 7.2 Weighted Overall Quality

$$\text{Quality} = 0.30(\text{pH}) + 0.20(\text{slurry}) + 0.25(\text{reagent}) + 0.25(\text{feed})$$

The weights sum to 100%, keeping the overall indicator within a 0–100% range.

### 7.3 Grade Classification

| Estimated Quality | Displayed Status |
|-------------------|------------------|
| ≥ 90%             | High Grade       |
| 75–89.99%         | Medium Grade     |

60–74.99%

Low Grade

< 60%

Very Low Grade

## 8. Technology Stack

| Technology | Purpose                           |
|------------|-----------------------------------|
| Python     | Application and calculation logic |
| Streamlit  | Interactive web dashboard         |
| Pandas     | Result and target tables          |
| Math       | Exponential scoring calculation   |

## 9. Application Features

- Interactive sidebar controls for four process parameters.
- Current process settings shown as dashboard metrics.
- One-click quality analysis.
- Estimated mineral-quality percentage.
- Grade/status classification.
- Individual parameter scores and weights.
- Target operating values.
- Progress bars for quick visual comparison.
- Diagnostic recommendations.
- Calculation explanation and prototype warning.

## 10. Development Approach

7. Defined process parameters and prototype operating targets.
8. Implemented a reusable closeness-score function.
9. Implemented weighted quality calculation.
10. Added status classification and recommendation logic.
11. Built the Streamlit dashboard.
12. Added tables, progress indicators and documentation.
13. Clearly separated prototype assumptions from validated measurements.

## 11. Testing Strategy

| Test Case                | Expected Behaviour      | Purpose                  |
|--------------------------|-------------------------|--------------------------|
| All values at targets    | Quality reaches 100%    | Verify optimum condition |
| One parameter moved away | Overall score decreases | Verify sensitivity       |

|                               |                                 |                                |
|-------------------------------|---------------------------------|--------------------------------|
| Several parameters moved away | Overall score decreases further | Verify weighted combination    |
| Extreme allowed values        | Score remains 0–100%            | Check robustness               |
| Same inputs repeated          | Same result                     | Verify deterministic behaviour |

Software behaviour can be tested directly; physical validity requires comparison with laboratory or plant measurements.

## 12. Current Limitations

- No live sensor or plant-control connection.
- No historical production dataset in the current version.
- Targets, tolerances and weights are prototype assumptions.
- Output is an estimated score, not a laboratory assay.
- Complex interactions between variables are not explicitly modelled.
- The CSV-free version has not been statistically validated against labelled quality measurements.

## 13. Future Development

- Collect synchronized process parameters and laboratory quality measurements.
- Calibrate targets, tolerances and weights.
- Compare the rule-based estimator with regression and machine-learning models.
- Add historical trend charts and time-series analysis.
- Integrate live sensor/API data.
- Add anomaly detection and early-warning alerts.
- Add authentication, site configuration and audit trails.
- Deploy the application to a secure server/cloud environment.

## 14. Interview Explanation

### 30-second project explanation

“I developed a Streamlit-based mining quality-control prototype that estimates a mineral-quality indicator from four process parameters: pH, slurry intensity, reagent dose and feed rate. I implemented a transparent weighted scoring method that compares each input with a target operating region and combines the parameter scores into an overall indicator. The dashboard also shows parameter contributions and diagnostic recommendations. The current version is a prototype; my next step would be to calibrate and validate it using laboratory or plant data and then compare it with a machine-learning approach.”

### Questions you should be ready for

#### Why Streamlit?

It allowed me to build an interactive engineering dashboard directly in Python without developing a separate frontend.

### **Why not ML yet?**

A validated dataset is needed before a machine-learning model can be meaningfully trained and evaluated. The first version is intentionally transparent and CSV-free.

### **What does the percentage mean?**

It is a calculated prototype quality indicator, not a direct physical measurement.

### **Why use weights?**

Different parameters can have different importance, so weights represent their relative contribution in the prototype.

### **How would you improve it?**

I would collect validated data, calibrate the model, evaluate error metrics, quantify uncertainty, and integrate live process data.

## **15. Resume / LinkedIn Description**

- Developed a Streamlit-based mining quality-control prototype for estimating a mineral-quality indicator from four process parameters.
- Implemented a transparent weighted scoring algorithm with parameter-level diagnostics and configurable operating targets.
- Built an interactive dashboard with process controls, quality status, contribution tables, progress indicators and recommendations.
- Designed the prototype so the rule-based calculation can later be calibrated or replaced with validated statistical or machine-learning models.

## **16. Project Deliverables**

- Python application: mineral quality prototype 1.py
- Desktop launcher batch file for starting the local Streamlit application
- Professional project documentation
- Future deliverables: validated dataset, calibration report and model-comparison report

## **17. Conclusion**

The project demonstrates an end-to-end engineering software workflow: identifying a process-monitoring problem, defining measurable inputs, implementing a transparent calculation method, developing an interactive interface, and documenting assumptions and limitations. The key next step is validation against real laboratory or plant measurements so the prototype can progress from a demonstrator into a defensible engineering decision-support system.